

ZAIDAN KHAN

Embedded Hardware Engineer | Hardware Design Engineer | Embedded R&D Engineer
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PROFESSIONAL SUMMARY

B.Tech ECE graduate and Embedded R&D Engineer with 6+ months of hands-on embedded hardware design experience. Proficient in multi-sheet schematic capture (Altium Designer), PCB layout, signal integrity, impedance matching, component selection, BOM preparation, vendor sourcing, and datasheet analysis for safety-critical medical and automotive-grade systems. Experienced in power electronics, RF/analog subsystem integration, EMC/EMI-aware layout, motor drive circuits, battery management systems (BMS), and IEC 62304-compliant firmware on STM32 ARM Cortex-M4. Adept at cross-functional sprint-based delivery and design-for-manufacture (DFM). Targeting Embedded Hardware Engineer, Hardware Design Engineer, or Embedded R&D Engineer roles.

EDUCATION

B.Tech — Electronics & Communication Engineering | Technocrats Institute of Technology (Excellence), Bhopal | CGPA: 7.79/10 | Graduated June 2026

TECHNICAL SKILLS

Hardware Design: Schematic Capture, Multi-sheet PCB Design, Signal Integrity, Impedance Matching, Net Classes, DRC/ERC, Layout Review, Gerber Generation, Design for Manufacture (DFM)

EDA & Simulation: Altium Designer, KiCad, EasyEDA, Proteus, LTspice (analog circuit simulation & power rail verification)

Power & Analog: Buck/Boost Converters, Li-ion BMS, Battery Charger ICs, UPS Circuitry, Automotive 12/24V Systems, Thermal Management, Decoupling & Bypass Capacitor Design, ESD & Overcurrent Protection

RF & Wireless: Antenna Matching Networks, SAW/BAW Filtering, TCXO Clocking, LNA, EMC/EMI Layout, LTE/4G/GNSS/Wi-Fi/BT RF Integration

Component Engineering: BOM Preparation, Datasheet Analysis, Component Selection, Footprint Verification, Vendor Sourcing, India-specific Sourcing, Alternate Part Cross-referencing

MCU & Firmware: STM32 ARM Cortex-M4, ESP32/ESP32-S3, AVR; Embedded C/C++, FreeRTOS, UART/SPI/I2C/CAN Bus, MQTT, LVDS

Debug & Version Control: GDB, OpenOCD, Git, JTAG/SWD, Oscilloscope, Logic Analyzer, Serial Console Tracing, Keil uVision, STM32CubeIDE

INTERNSHIP EXPERIENCE

Embedded R&D Engineer (Intern) | Walnut Medical, Mohali, Punjab | Jan 2026 – Jul 2026

Primary — Smart Infusion Pump (STM32 ARM Cortex-M4, FreeRTOS, Altium Designer):

- Designed full 5-sheet Altium multi-sheet schematic (Power, MCU/Safety, Motor Drive (DRV8825 stepper), Sensors/UI/Battery, Peltier/ESP32); applied net classes, DRC/ERC, and schematic review across all sheets.
- Identified and corrected critical hardware errors: BQ24650 VFB feedback divider miscalculation, MPRLS I2C/SPI symbol mismatch, ATmega328P crystal (16 → 8 MHz for 3.3 V operation), I2C pull-up values, and net label mismatches.
- Redesigned battery architecture from 2S to 3S2P (Samsung INR18650-30Q) with BQ24650 charger, BQ76920 AFE, XL6009E1 UPS boost converter, and decoupling network for power rail stability.
- Prepared 2-layer PCB (~100×120 mm): placement planning, thermal management zones, routing constraints, DRC validation, and Gerber generation for manufacturing; conducted India-specific vendor sourcing and BOM cross-referencing.
- Architected FreeRTOS firmware across 6 concurrent tasks (sub-10 ms response); PID closed-loop flow control at ±2% infusion accuracy; IEC 62304-compliant design documentation across all sprint milestones.

Secondary — Automotive Telematics Gateway (4G LTE + GNSS + IoT, ESP32-S3, Altium Designer):

- Designed automotive GPS gateway hardware integrating ESP32-S3 MCU, GW1NS LTE module, BK1616P GNSS receiver, and Wi-Fi/BT interfaces; completed multi-sheet Altium schematic with datasheet analysis, BOM preparation, and footprint verification.
- Designed 12/24V automotive power system: OC2006 buck converter, BQ24074 Li-ion charger, battery backup, reverse-polarity and surge protection, EMC-aware decoupling strategy.
- Integrated LTE/GNSS/Wi-Fi RF front-end: antenna matching networks, TCXO clocking, SAW filtering, LNA path, and ESD protection for automotive EMC/EMI compliance.
- Completed full PCB design workflow: signal integrity planning, placement, routing constraints, DRC validation, and Gerber generation for manufacturing hand-off.

Technical Trainee | BHEL (Bharat Heavy Electricals Ltd.), Bhopal | Aug – Sep 2025 (2 Weeks)

- Multilayer PCB design workflows (stack-up planning, DRC in Altium), signal conditioning for EV sensor interfaces, and CAN bus ECU-to-actuator communication mapping.

INDEPENDENT PROJECTS

Adaptive Traffic Light Controller | ARM Cortex-M4, Embedded C, IR Sensors, PWM, Proteus

- ISR-driven density-based signal control for 4 intersection lanes; ~22% lower simulated average wait time vs. fixed-cycle baseline; flash-stored configurable thresholds, PWM actuation.

Smart Home Automation System | ESP32, C++, MQTT, IoT, Blynk, DHT11, BMP280

- ESP32 + MQTT IoT system monitoring 3 environmental parameters (temperature, humidity, pressure); Blynk remote control; end-to-end latency under 500 ms.

CERTIFICATIONS & TRAINING

- PCB Design and EV Embedded Systems — BHEL Bhopal (Sep 2025)
- Altium Designer — Altium Academy (PCB design rules, signal integrity, DFM guidelines)
- FreeRTOS Real-Time Kernel — FreeRTOS.org + Mastering the FreeRTOS Real Time Kernel (Richard Barry)
- ARM Cortex-M Architecture and Programming — ARM official documentation and Cortex-M4 Technical Reference Manual